Check for updates

JOURNAL OF COMPUTATIONAL BIOLOGY
Volume 32, Number 6, 2025
© Mary Ann Liebert, Inc.
Pp. 558–572
DOI: 10.1089/cmb.2024.0631

Original Articles

Open camera or QR reader and scan code to access this article and other resources online.

FedOpenHAR: Federated Multitask Transfer Learning for Sensor-Based Human Activity Recognition

EGEMEN İŞGÜDER and ÖZLEM DURMAZ İNCEL

ABSTRACT

Wearable and mobile devices equipped with motion sensors offer important insights into user behavior. Machine learning and, more recently, deep learning techniques have been applied to analyze sensor data. Typically, the focus is on a single task, such as human activity recognition (HAR), and the data is processed centrally on a server or in the cloud. However, the same sensor data can be leveraged for multiple tasks, and distributed machine learning methods can be employed without the need for transmitting data to a central location. In this study, we introduce the FedOpenHAR framework, which explores federated transfer learning in a multitask setting for both sensor-based HAR and device position identification tasks. This approach utilizes transfer learning by training task-specific and personalized layers in a federated manner. The OpenHAR framework, which includes ten smaller datasets, is used for training the models. The main challenge is developing robust models that are applicable to both tasks across different datasets, which may contain only a subset of label types. Multiple experiments are conducted in the Flower federated learning environment using the DeepConvLSTM architecture. Results are presented for both federated and centralized training under various parameters and constraints. By employing transfer learning and training task-specific and personalized federated models, we achieve a higher accuracy (72.4%) compared to a fully centralized training approach (64.5%), and similar accuracy to a scenario where each client performs individual training in isolation (72.6%). However, the advantage of FedOpenHAR over individual training is that, when a new client joins with a new label type (representing a new task), it can begin training from the already existing common layer. Furthermore, if a new client wants to classify a new class in one of the existing tasks, FedOpenHAR allows training to begin directly from the task-specific layers.

Keywords: federated transfer learning, multi-task learning and human activity recognition.

Faculty of EEMCS, Pervasive Systems Research Group, University of Twente, Enschede, The Netherlands.

1. INTRODUCTION

Motion sensors integrated into wearable and mobile devices are commonly used to track users' habitual patterns, including the activities they perform, their locations, and where they carry their devices. In particular, monitoring physical activity levels can help individuals stay active and reduce the risk of chronic diseases, such as obesity. Machine learning and, more recently, deep learning techniques have been applied to sensor data for activity recognition and other related tasks (Kumar et al., 2024). While there are efforts to run learning algorithms directly on devices, wearables are still limited in terms of computational power and memory, so processing is typically performed on a central server or in the cloud. Another concern is that centralizing private data for processing may infringe on user privacy (Truong et al., 2021), which has prompted the exploration of federated learning as a potential solution.

Federated learning is a novel machine learning technique introduced by Google (McMahan et al., 2017). Its primary goal is to process data while safeguarding privacy. With the growing emphasis on data privacy, particularly following the updates to the General Data Protection Regulation, federated learning offers a promising approach. Instead of sharing the actual data among participants (clients), federated learning shares only model parameters, enabling distributed learning. This method enhances privacy by reducing the transmission of sensitive data and alleviates the resource burden on smaller devices, such as wearables and IoT devices, by limiting the need for data transmission.

Federated learning is especially valuable in applications like sensor-based human activity recognition (HAR). Motion sensors, including accelerometers and gyroscopes, capture data that is then processed using machine learning techniques to monitor user activities, such as walking or sitting, as well as more specialized activities, such as sports or emergency situations, such as falls.

Federated learning is mostly employed using a single source of data and then splitting the source into multiple "slices" to obtain a scheme where each slice is then dedicated to a participant of the federated learning approach, referred to as "clients" (Yang et al., 2019). However, what would happen if those slices differ not only by size but also come from completely different sources? Then, a new challenge arises as the heterogeneity of the slices of the data becomes a key point to consider. To our knowledge, that type of heterogeneity of data sources has not been examined before. In our research, we wanted to determine how well the federated learning method would perform under the data heterogeneity scenario. For that purpose, we focus on recognizing two types of tasks: human activities and sensor device positions. For both tasks, we used the DeepConvLSTM model, which is proven to perform well with tabular HAR data by the authors in Ordóñez and Roggen (2016). During the experiments, the data from the OpenHAR framework (Siirtola et al., 2018) is used, which consists of ten smaller datasets. Together, it encompasses accelerometer sensor data located in various parts of the body, gathered from 211 participants. In this article, we present a new federated multitask learning architecture called FedOpenHAR, *combining federated learning and multitask learning*, and we further evaluate FedOpenHAR with two different approaches: one being called *centralized learning*, representing no data privacy and putting all data into one big pile, and the other approach being *individual learning*, in which no data is shared, conserving privacy, where all participants do their own training and do not communicate in any way.

As mentioned, the tasks to learn during the experiments are the users' activities and the devices' positions. To the best of our knowledge, *this study is the first to implement a solution to a multitask classification problem within multiple different datasets employing a federated transfer learning method, where all datasets do not contain every type of label.*

The article continues as follows: Section 2 examines the recent works in the area, Section 3 explains the used dataset in experiments, and Section 4 explains the selected classification algorithm and the federated learning architecture. Section 5 focuses on the results of experiments, and Section 6 summarizes the results obtained and opens a path for further studies.

2. RELATED WORK

Federated learning has gained significant popularity in the research community since its introduction by Google in 2017 (McMahan et al., 2017). In their article, the authors define the concept of federated learning and present the first federated averaging algorithm, FedAVG. They also perform experiments and report their results using the CIFAR-10 dataset.

The study in Yang et al. (2019) offers a comprehensive technical overview of federated learning implementation, illustrating how it can be integrated with various machine learning approaches. It distinguishes between horizontal federated learning (where data features are consistent) and vertical federated learning (where data sources are the same), providing valuable insights into these methods. However, the article does not explore federated transfer learning (where both data sources and features vary), which remains an open area for future research. In our study, in addition to the differences in both data sources and features, we also consider variations in **task types** (e.g., activity name/sensor position) and **label types for each task** (such as running, walking, or falling), which differ across clients.

In this study, our primary application is sensor-based HAR, a topic that has been widely explored in the literature. One of the pioneering works on federated learning for sensor-based HAR is Chen et al. (2020), which introduces a federated learning platform for wearable healthcare devices. The study uses the UCI (University of California Irvine) public smartphone dataset (Anguita et al., 2013) to train models. This dataset is a subset of the OpenHAR-integrated datasets used in our research. Unlike our study, which employs deep learning techniques, FedHealth (Chen et al., 2020) utilizes machine learning approaches such as K-nearest neighbors and support vector machines.

Numerous studies employing federated learning to HAR compare it with centralized approaches. For instance, in Kalabakov et al. (2023), the authors compare centralized deep learning with federated learning for activity classification. They find that both techniques yield similar results in terms of macro F1-scores and confusion matrices. While federated learning can match the effectiveness of centralized learning, it offers the added benefit of enhanced privacy.

Federated learning relies on aggregating model parameters, choosing an averaging algorithm critical for its success. In Ek et al. (2020), different aggregation techniques, such as FedPer, FedMA, and FedAVG, were assessed using HAR data, and their performances were compared. The findings indicate that the baseline FedAVG algorithm outperformed the others. The additional steps in FedMA and FedPer aim to enhance the personalization of local models; however, this increased focus on personalized models negatively impacted their performance on the global test set, resulting in lower accuracy.

Clients (participants) in federated learning vary in technical capabilities, data composition, and feature distribution, a challenge known as heterogeneity. This heterogeneity significantly impacts the effectiveness of federated learning. Personalizing the aggregated model for each client is a promising approach to address this issue. In Arivazhagan et al. (2019), personalization is explored through a novel algorithm called FedPer, which customizes the final layer of each client's model. This is achieved using a specialized federated averaging algorithm combined with additional computations based on stochastic gradient descent.

Further studies on personalization in federated learning include Wu et al. (2020) and Tan et al. (2023). The authors of Wu et al. (2020) identify limitations of FedAVG when dealing with highly skewed data and propose a personalized version of FedAVG. Meanwhile, Tan et al. (2023) leverages transfer learning for personalization, utilizing FedMD and FedStag as aggregation algorithms to enhance performance.

The recent trend of integrating multitask learning with federated learning is exemplified in Smith et al. (2018). The authors introduce a framework called MOCHA, which enables simultaneous optimization of linear and nonlinear learning tasks. Their experiments utilize three diverse datasets, including a HAR dataset where each individual is treated as a different task.

One of the most relevant studies, addressing diverse aspects such as multitask learning, hierarchical layering, and transfer learning, is presented in Keçeci et al. (2022). The authors introduce a novel federated learning framework that integrates multitask, transfer, and federated learning methodologies. Their proposed architecture comprises four distinct layers: pretrained, common, task-specific, and personalized layers. The experiments are conducted on a synthetically generated network using binary-labeled face image datasets, categorizing images into two distinct classes. Their approach achieves results comparable to the baseline method, which involves training models individually.

A key difference between their study and ours lies in the strategy for implementing the data distribution. In their work, a single dataset is partitioned among clients, whereas in our study, each client operates on entirely separate datasets, which may or may not encompass all available labels. Additionally, their study focuses on image recognition tasks, while our work targets sensor-based HAR, employing different baseline deep learning architectures. Furthermore, while Keçeci et al. (2022) deals with binary classification tasks, our study addresses multi-class label classification.

To date, federated multitask learning has been employed for various tasks in the image recognition domain, and federated transfer learning has been applied in the HAR domain. However, this research is the first to implement sensor-based HAR implemented with a federated multitask transfer learning architecture. The comparison of different aspects of this article with the mentioned studies can be seen in Table 1.

In Table 1, the columns represent different properties of the mentioned studies, with many of them abbreviated due to the size constraints of the table. The columns of the table can be named respectively: "*Centralized Learning,*" "*Federated Learning,*" "*Multi-Task Learning,*" "*Transfer Learning,*" "*Personalization,*" "*Human Activity Recognition,*" "*Different Dataset for Each Client,*" "*Different Labels for Each Client*" and "*DeepConvLSTM*."

3. DATASET

As mentioned, OpenHAR is actually an ensemble of ten different datasets. They are all collected within different research studies and with different participants. The OpenHAR subsets include: dataset_ID (integer between 1–10), position_ID (integer between 1–14), user_ID (integer between 0–66), activity_ID (integer between 1–99), timestamp, x–y-z and magnitude acceleration values. We focused on HAR and position recognition, i.e., where the phone was carried, tasks. Hence, from these columns, x, y and z axis and magnitude accelerations are used as features during training, whereas activity_ID and position_ID are used as the labels.

3.1. Preprocessing

Preprocessing of labels is necessary for the following main reasons:

- There were quite similar labels (e.g. sitting vs. sitting in car) which are hard or not very meaningful to classify,
- Some labels had very few instances (e.g. walking at stairs),
- Some labels existed only in one or two datasets (e.g. elevator up and elevator down).

OpenHAR includes 16 different activity labels in its original version. During the preprocessing of activity labels; NULL labels are deleted; Standing, Sitting, Idling, Sitting in Car, Elevator Up and Elevator Down labels are combined into Idling; Walking, Walking stairs up, Walking stairs down, Walking at stairs (inc. up and down) are combined into Walking. In Figure 1, the labels of activities of OpenHAR can be seen before and after the preprocessing. OpenHAR comprises 14 distinct position labels in the unprocessed version. During the data collection of some datasets, sensor position was defined without any direction (left or right), meaning that it can either be left or right, while in some cases, the position was explicitly defined as left or right (denoted by *L* or *R*). In addition, in some cases, study subjects were allowed to decide the orientation of the sensor, while in some studies, orientation was fixed, which is denoted by *Free* or *Fixed*. During the preprocessing steps; Hip, Trouser's pocket left (fixed orientation), Trouser's pocket right (fixed orientation), Trouser's pocket left (free orientation), Trouser's pocket right (free orientation) are all combined into Pocket; Shin, Ankle, Foot left, Foot right are all combined into Leg/Foot. The reason for re-combinations is either they have very few

TABLE 1. COMPARISON WITH RELATED STUDIES

	Properties								
Articles	*Centr*	*Fed*	*MT*	*TL*	*Per*	*HAR*	*Set/cli*	*Lab/cli*	*DCL*
Keçeci et al. (2022)	X	X	X	X	X				
Chen et al. (2020)	X	X		X	X	X			
Kalabakov et al. (2023)	X	X				X			
Arivazhagan et al. (2019)		X	X		X				
Ouyang et al. (2021)	X	X		X		X			
Presotto et al. (2022)		X		X	X	X			
Smith et al. (2018)		X	X			X			
Bursa et al. (2023)	X			X		X			X
Our Study	X	X	X	X	X	X	X	X	X

Centr, centralized learning; DCL, deepConvLSTM; Fed, federated learning; HAR, human activity recognition; Lab/cli, different labels for each client; MT, multi-task learning; Per, personalization; Set/cli, different dataset for each client; TL, transfer learning.

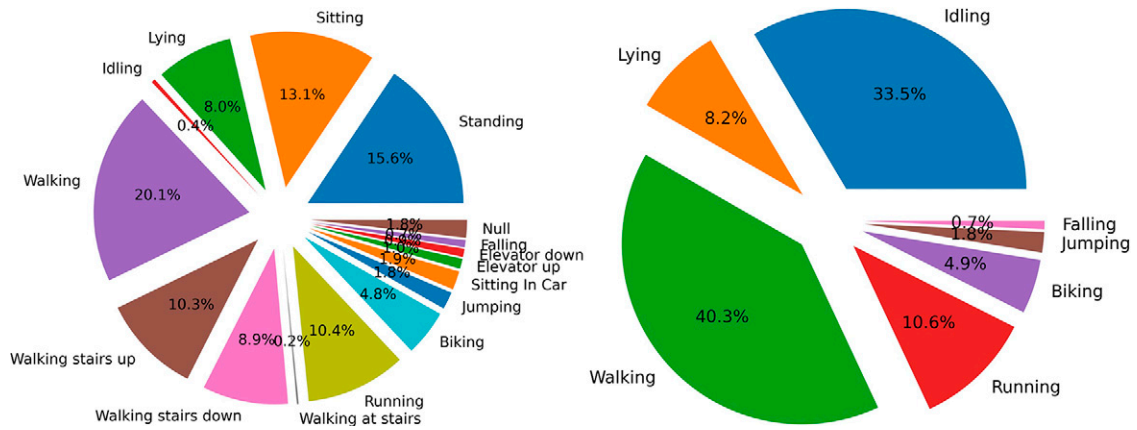

FIG. 1. Activity label distribution of OpenHAR, before(left)/after(right) preprocessing. HAR, human activity recognition.

instances, or they are strongly correlated positions, from almost the same body parts. In Figure 2, the labels of activities of OpenHAR can be seen before and after the preprocessing. Summarized information about the datasets such as sampling rates as well as unique label amounts (after the preprocessing) are presented in Table 2.

Although the 10 different datasets were collected at different sampling rates, OpenHAR synchronized each dataset to have the same sampling rate at 10 Hz. During our experiments, the 10 datasets that are mentioned in Table 2, with 10 different Dataset_IDs, are used both for their activity and position labels, which we consider as 10 activity datasets and 10 position datasets (the features are the same, label&label types are different), thus allowed us to allocate them into 20 different clients.

However, between the 10 position datasets, 5 of them contained only one unique position label (meaning they were collected from just one body position) and 2 of them contained only two different position labels (meaning they were collected from only two different body positions, and heavily unbalanced). That is the reason that 7 of the position datasets were excluded from our experiments. In the end, we obtained 13 datasets, 10 of them being for activity and 3 of them being for position labels.

4. FEDOPENHAR

In this section, we highlight the details of FedOpenHAR, its layered hierarchy, and the performance metrics used in the evaluations. FedOpenHar is a federated multitask transfer learning approach inspired by the architecture of Keçeci et al. (2022), which is designed for image recognition tasks. The main layered hierarchy can

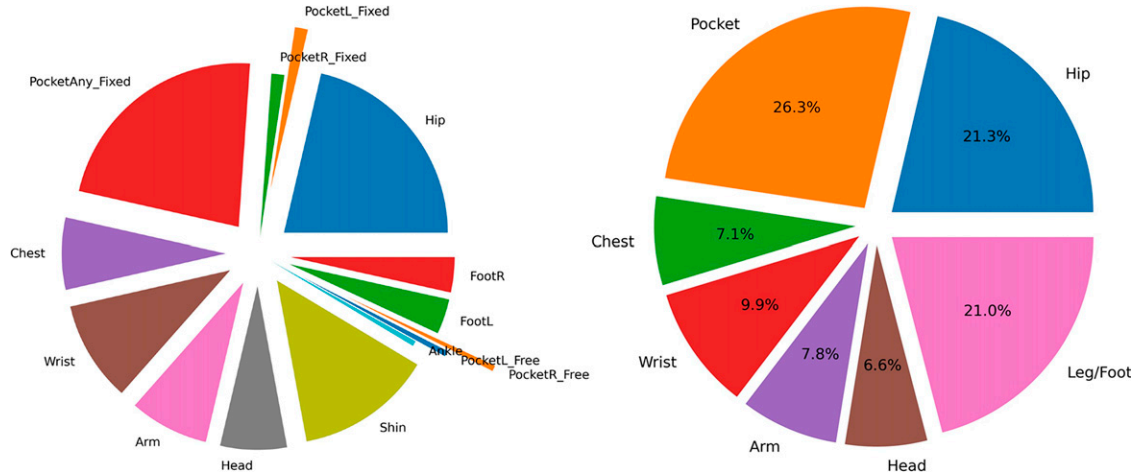

FIG. 2. Position label distribution of OpenHAR, before(left)/after(right) preprocessing.

TABLE 2. DETAILS OF 10 DATASETS OF OPENHAR

Dataset ID	*Dataset name*	*Unique activity amount*	*Unique position amount*	*Sampling frequency*
1	Banos et al.	6	3	50 Hz
2	Ortiz et al.	3	1	50 Hz
3	Shoaib et al.	4	4	50 Hz
4	Siirtola et al.	4	1	40 Hz
5	Stisen et al.	3	2	50–200 Hz
6	USC-HAD	5	1	100 Hz
7	UniMib-SHAR	4	1	50 Hz
8	HuGaDB	4	2	60 Hz
9	RealWorldHAR	5	7	50 Hz
10	MobiAct	6	1	200 Hz

be seen in Figure 3. Whenever a layer's training concludes, its parameters are frozen, and the training is applied to the following layers.

The action flow of FedOpenHAR's process is depicted in Figure 4 and can be summarized as follows:

(i) Server Selection: A client is designated to function as the server.
(ii) Client Connections: Clients connect to the server via simulation or port-socket connections (Step 1 in Fig. 4). Each client focuses on classifying one task at a time, but they are free to choose their tasks, allowing for a multitask setup.
(iii) Weight Initialization (Pretraining): At the start of the algorithm, the server initializes the model weights (Step 2 in Fig. 4), forming the pretrained layers.
(iv) Training Common Layers:

- Client-Side Training: Clients use the same initialized weights to train their models on local data (Step 3 in Fig. 4). This training runs for a specified number of per-client epochs.
- Aggregation: After training, clients send their model parameters to the server for aggregation (Step 4 in Fig. 4). Details of the aggregation process are discussed in the next section.
- Global Model Update: The server sends the aggregated global model back to the clients (Step 5 in Fig. 4), and the training process repeats iteratively (Steps 3-4-5) for a predefined number of communication rounds.

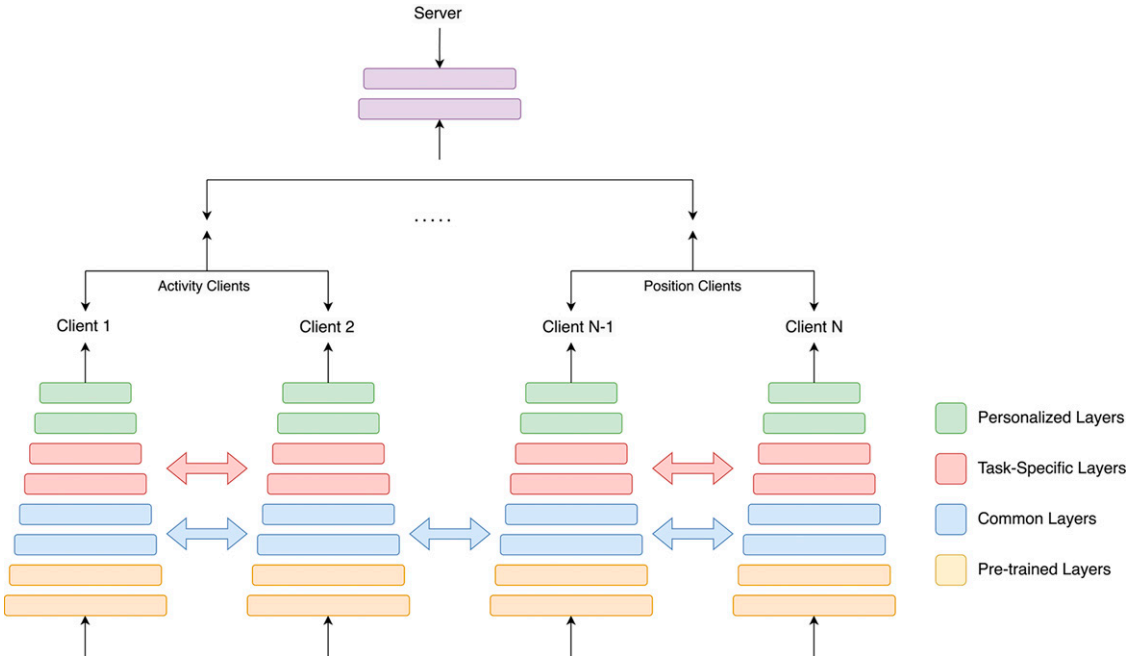

FIG. 3. Layered hierarchy of FedOpenHAR.

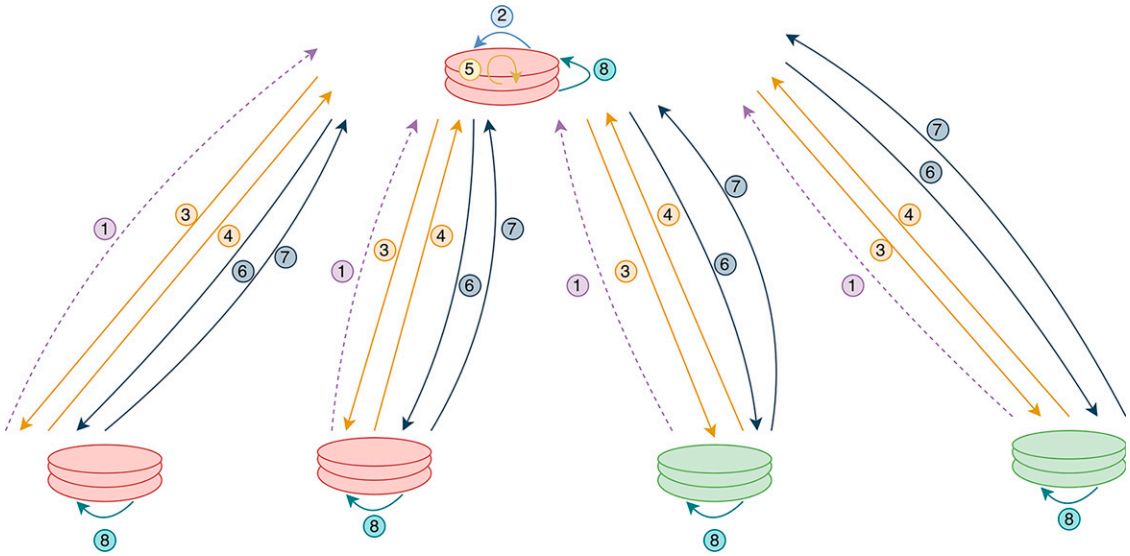

FIG. 4. Algorithmic flow of FedOpenHAR.

(v) Training Task-Specific Layers:

- Task-Specific Training: For each task, clients start with the common layer weights and train locally on task-specific data (Step 6 in Fig. 4) for a defined number of epochs.
- Task-Specific Aggregation: Parameters are sent to the server for aggregation (Step 7 in Fig. 4).
- Global Model Update for each Task: The global model is updated and redistributed to clients (Step 5 in Fig. 4). This process (Steps 5-6-7) repeats for a set number of communication rounds.

(vi) Personalization: In the final phase, each client locally fine-tunes the global model obtained after task-specific training. This personalization step, performed without further federation, completes after the clients train the model for a defined number of epochs (Step 8 in Fig. 4).

(vii) Completion: All clients now possess unique and personalized models.

Throughout the training, a "test" function is periodically invoked to evaluate training progress. On the client side, SGD (Stochastic Gradient Descent) is the default optimizer, and Cross-Entropy Loss is used as the criterion for training.

4.1. DeepConvLSTM architecture

In the experiments, the DeepConvLSTM architecture Ordóñez and Roggen (2016) is employed as the primary model. This model integrates a Deep Convolutional Neural Network with an LSTM (Long Short-Term Memory) network. Its architecture includes four convolutional layers, followed by two dense layers, and concludes with a softmax classifier layer. The convolutional layers serve as feature extractors, generating feature maps from the input tabular data, while the recurrent layers capture and model the temporal dynamics of these feature maps.

DeepConvLSTM is chosen due to its demonstrated effectiveness in handling tabular data, particularly in the domain of HAR, as highlighted in Ordóñez and Roggen (2016). The model has also been utilized in various HAR-related studies. For instance, in Murahari and Plötz (2018), the authors enhance DeepConvLSTM with attention mechanisms to improve activity recognition performance. Similarly, in Bock et al. (2021), the architecture is employed to advance deep learning techniques in HAR applications. The algorithmic workflow of DeepConvLSTM is illustrated in Figure 5.

From the left, the signals coming from the wearable sensors are processed by four convolutional layers, which allow learning features from the data. Two dense layers then perform a non-linear transformation, which yields the classification outcome with a softmax logistic regression output layer on the right. Input at Layer 1 corresponds to sensor data of size $D \times S^1$, where D denotes the number of sensor channels and S^1 the length of features maps in layer l. Layers 2–5 are convolutional layers. K^1 denotes the kernels in layer l (depicted as red squares). F^1 denotes the number of feature maps in layer l. In convolutional layers, a_i^l denotes the activation that defines the feature map i in layer l. Layers 6 and 7 are dense layers. In dense layers, $a_{t,I}^l$ denotes the

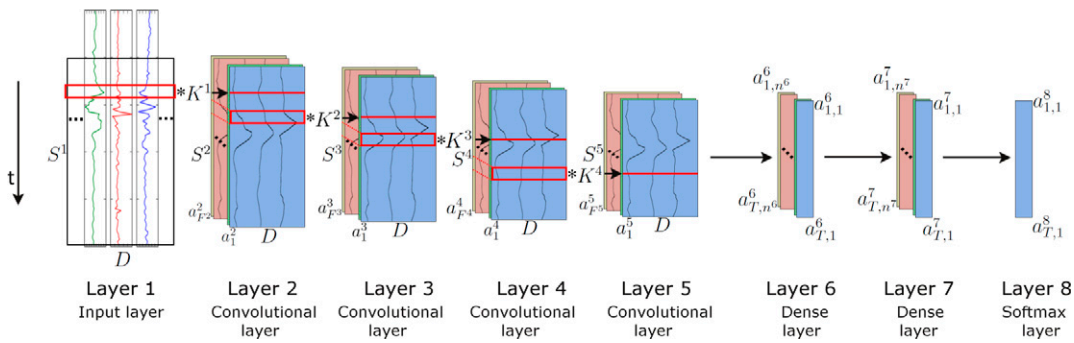

FIG. 5. Architecture of the DeepConvLSTM framework for activity recognition Ordóñez and Roggen (2016).

activation of the unit i in hidden layer l at time t. The time axis is vertical. Figure 5 and the network's flow explanation are taken from the study of Ordóñez and Roggen (2016).

5. EXPERIMENTS

We performed different experiments to evaluate the performance of the techniques mentioned above individually and in various combinations. First, all data is combined, and training is performed in a centralized manner. Second, every client's data is trained individually, without any parameter sharing, and then simple federated learning for one task and multitask training are explored. Finally, the results of the multitask federated transfer learning experiments with FedOpenHAR are presented. For all experiments, data is always split into train and test with 80%–20% ratio.

5.1. Metrics

The default success metric used in all experiments is **accuracy**. Accuracy for each client i is calculated as in Equation 1 (TN: True Negative, TP: True Positive, FN: False Negative, FP: False Positive):

$$Acc_i = \frac{\text{Count of all correctly classified instances}}{\text{Count of all instances}} = \frac{TP+TN}{TP+TN+FP+FN} \tag{1}$$

In experiments involving multiple clients, relying on a single accuracy metric may not adequately represent the overall success of an algorithm. To address this, an additional metric, such as weighted average accuracy, is often introduced. Weighted average accuracy ($WeAvgAcc$) is calculated by assigning weights to the accuracy contributions of individual clients, as in Equation 2 (i denotes the id of each client, N denotes the total number of clients, K_i denotes the total number of instances in client i) :

$$WeAvgAcc = \frac{\sum_{i=1}^{N}(Acc_i \times K_i)}{\sum_{i=1}^{N} K_i} \tag{2}$$

Equation (2) is used during the calculation of the success of the whole FedOpenHAR or for the calculation of the success of a layer. While conducting experiments with FedOpenHAR, obtaining a comprehensive success metric poses significant challenges. The process involves the following steps:

- Client Test Accuracy: Each client records its test accuracy at the end of every training epoch.
- Selection of Best Accuracy: If a client's training spans multiple epochs, the highest accuracy among all epochs is considered as the accuracy metric for that client.
- Each client's accuracy is aggregated using a weighted average formula (as defined in Equation (2), which yields the overall average accuracy for a specific layer.
- This process is repeated for all layers, and the weighted average accuracy of the personalized (last) layer represents the overall performance of FedOpenHAR.

TABLE 3. CENTRALIZED APPROACH ACCURACIES

	Label type	
Learning type	*Activity*	*Position*
Centralized	69.8%	59.2%

As there are many steps of averaging for layers of FedOpenHAR, in order to keep the statistics simple and comparable, metrics like F1-score, precision, or recall are not used.

5.2. Centralized experiments

In centralized experiments, no federation tools are employed. Instead, traditional centralized structured learning steps, including training and testing, are implemented for each experiment.

During these experiments, data from all datasets within OpenHAR are combined into a single consolidated dataset. This approach disregards data privacy and directly contradicts the core objective of federated learning, which aims to preserve privacy by avoiding centralized data collection. In these experiments, classification is performed for each label type (activity and position) separately. The accuracy values achieved in these centralized experiments are presented in Table 3.

Although the CNN method performed well in activity labels, a dramatic drop can be observed in position labels, which, upon further examination of the confusion matrix, seemed due to the fact that the trained network favors one label as it was the most dominant label in all datasets combined.

5.3. Individual experiments

For the second class of experiments, instead of combining all data into one large dataset, each client (representing a dataset) is treated independently, using only its own data. This approach preserves data privacy but means that clients cannot benefit from the data of other clients. The position identification task is only trained

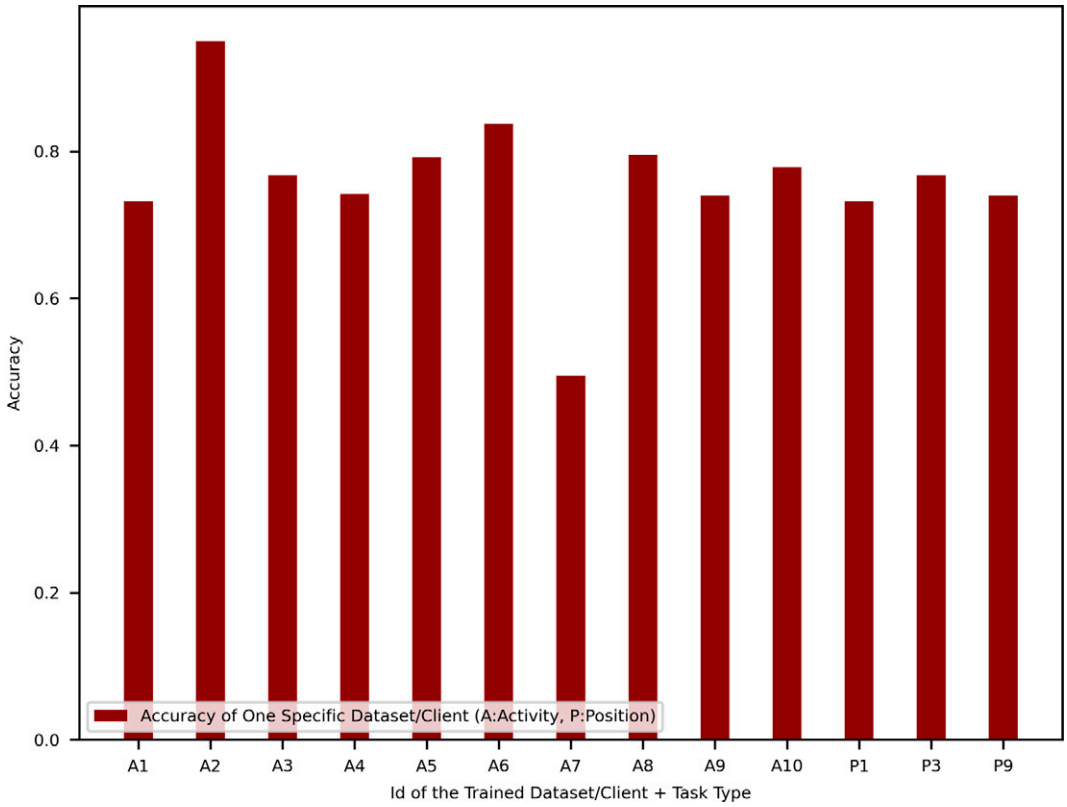

FIG. 6. Client-wise comparison of accuracies of individual clients' success.

TABLE 4. WEIGHTED ACCURACIES FOR FEDERATED EXPERIMENTS

	Label type	
Methods	*Activity*	*Position*
Federated one-task	76.7%	65.0%
Federated multitask	42.7%	

with three clients, as they are the only ones with more than two types of position labels in their data. The accuracy is reported as the weighted average accuracy across all clients and is used as the baseline for comparison with federated experiments. For all individual experiments, the data is split into training and testing sets using an 80%–20% ratio. These experiments are conducted using the best-performing network, DeepConvLSTM.

Figure 6 shows the performance with 13 datasets. These datasets consist of ten activity datasets and three position datasets. As mentioned, seven datasets are excluded from the position label classification task because they only contain one or two position labels, which are insufficient for effective classification. From the figure, we can observe the individual accuracy metric of each client. The first ten bars represent the activity clients, while the last three represent the position clients. The success of each client varies significantly depending on the dataset, with factors like the number of label types and dataset size influencing this variation. The weighted average accuracy across all clients is 72.6%, which serves as the baseline for comparison with the federated experiments.

5.4. Federated experiments

For the federated experiments, the setup mirrors the individual training experiments, but with the addition of model parameter sharing, which is the essence of federated learning. In this scenario, the ten datasets are referred to as "clients", and these clients do not share any private information from their datasets.

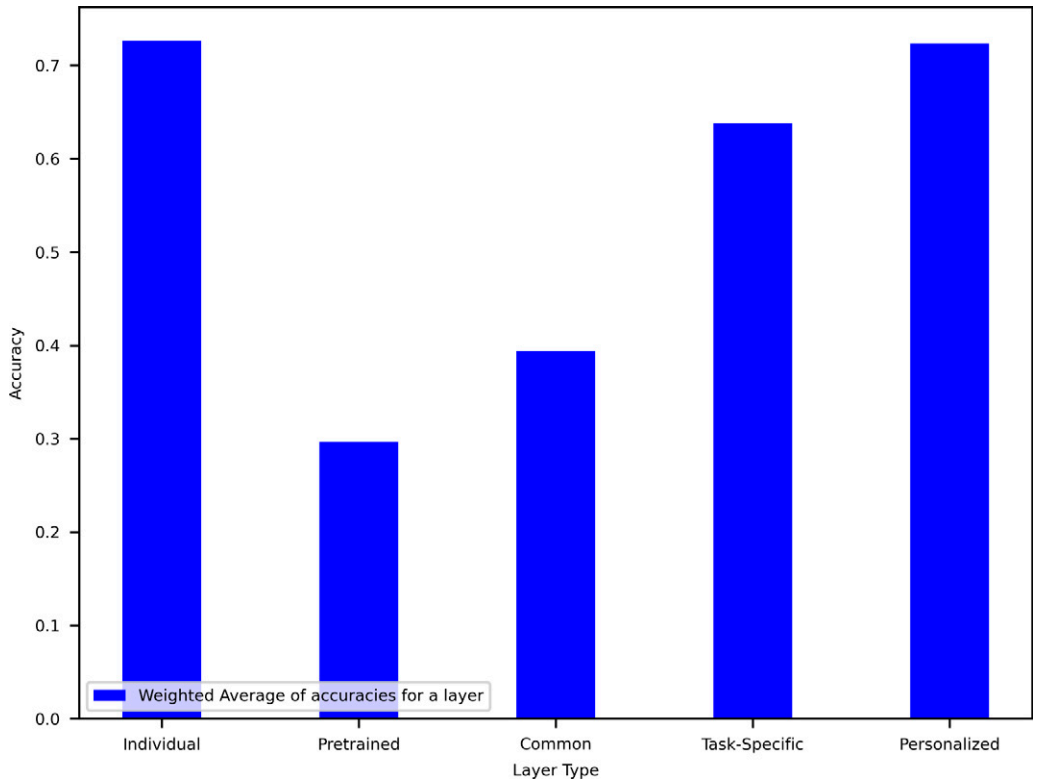

FIG. 7. Weighted average accuracies of clients by layers and comparison with the individual experiments' average.

For all federated experiments, the data is split into training and testing sets using a 90%–10% ratio. Each client performs only one training epoch on its device, and the simulations are run until ten communication rounds are completed, facilitating model aggregation after each round.

In the *federated one-task* experiments, each label type is trained separately. This means one simulation is conducted for the activity labels (using all ten clients), and another simulation is carried out for the position labels (using the three clients that have more than two position labels).

In the *federated multitask* experiments, both label types are learned simultaneously within the same experiment. Instead of running separate simulations for each label, activity and position labels are trained together in a single simulation. In this case, all 13 clients (activity and position clients combined) participate in the same federated simulation.

The accuracy drop from activities to positions is quite noticeable in Table 4., considering the minority of position clients. Another point to consider is that the multitask accuracy results are not sufficient at first glance; upon a deeper examination of their confusion matrices, the model favours some examples of labels more than others, which is not acceptable for proper learning. Overall the results are yet to approach centralized or individual results.

5.5. *Federated Multitask experiments with transfer learning (FedOpenHAR)*

In this section, we present the results of multitask transfer learning, which integrates federated learning, transfer learning, and multitask learning, as described in Section 4. The process begins with pretraining using one of the clients' data to initialize the layered approach. In this case, dataset-9 from OpenHAR for position labels achieved the best result. Following this, the next layer is trained using all 13 clients, resulting in the creation of the common layers.

Afterwards, two separate training sessions are conducted: one for the activity task (involving 10 clients) and another for the position task (involving 3 clients). These sessions yield the task-specific layers. Finally, each client fine-tunes its model separately, with the last layers trained using their own data. Before training each layer, the previous layers are frozen to retain their learned features.

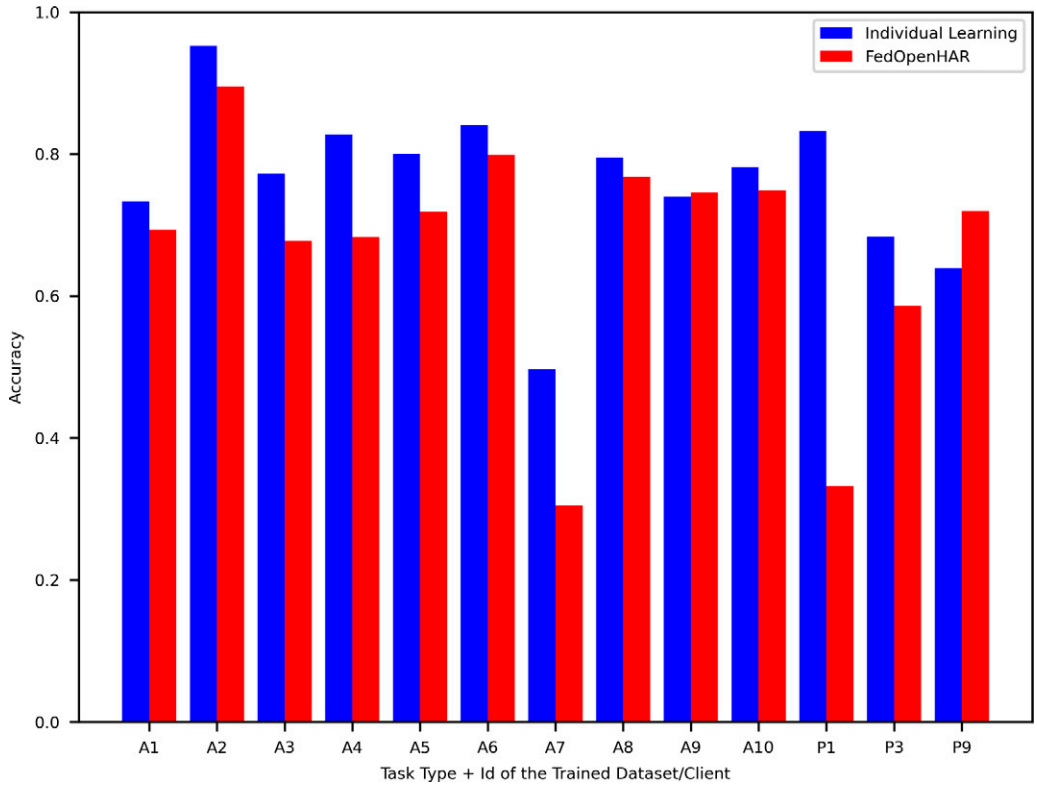

FIG. 8. Client-wise comparison of accuracies for individual learning and FedOpenHAR.

In Figure 7, a layer-wise comparison of the weighted accuracy of all clients for each layer is presented. This comparison also includes a reference to the individual setting. The final personalized layer achieved an accuracy similar to the baseline centralized setting, which is represented by the individual setting in this chart, as explained in Section 5.3).

In Figure 8, a client-wise comparison is shown, contrasting the individual experiments (explained in Section 5.3) with FedOpenHAR for all clients. The horizontal axis represents the clients, where the first letter indicates the task type (A for activity or P for position), and the number corresponds to the dataset ID. The vertical axis represents the accuracies in percentage.

The results appear quite comparable, but the scarcity of label types and the smaller dataset sizes negatively impact FedOpenHAR's success. Clients P1 and A7 show lower accuracy with FedOpenHAR compared to the individual (baseline) setting. This is mainly due to the fact that these two datasets are smaller in terms of instance size, and their label types are more limited than other datasets (as seen in Table 2).

The individual setting is less affected by these factors because it does not involve repeated averaging and weighting across multiple epochs, which is central to FedOpenHAR. This repeated process tends to disadvantage smaller datasets, as the model's ability to generalize across fewer data points is hindered.

Finally, the comparison of the performance of each layer in FedOpenHAR with the results from different experiments is shown in Figure 9. The figure displays the results of various techniques and compares them across different layers. FedOpenHAR outperforms both the centralized approach and the simple (unlayered) multitask approach, achieving nearly the same performance as the individual setting. The key advantage of FedOpenHAR becomes evident when new clients are introduced. FedOpenHAR can classify both activities and positions by default, and if a new client arrives with a completely new label type, it can begin training from the common layer. Additionally, if a new client needs to classify a new type of activity or position, it can start training directly from the task-specific layers. These advantages are not available in the individual setting, highlighting the flexibility and scalability of FedOpenHAR in handling new clients and tasks.

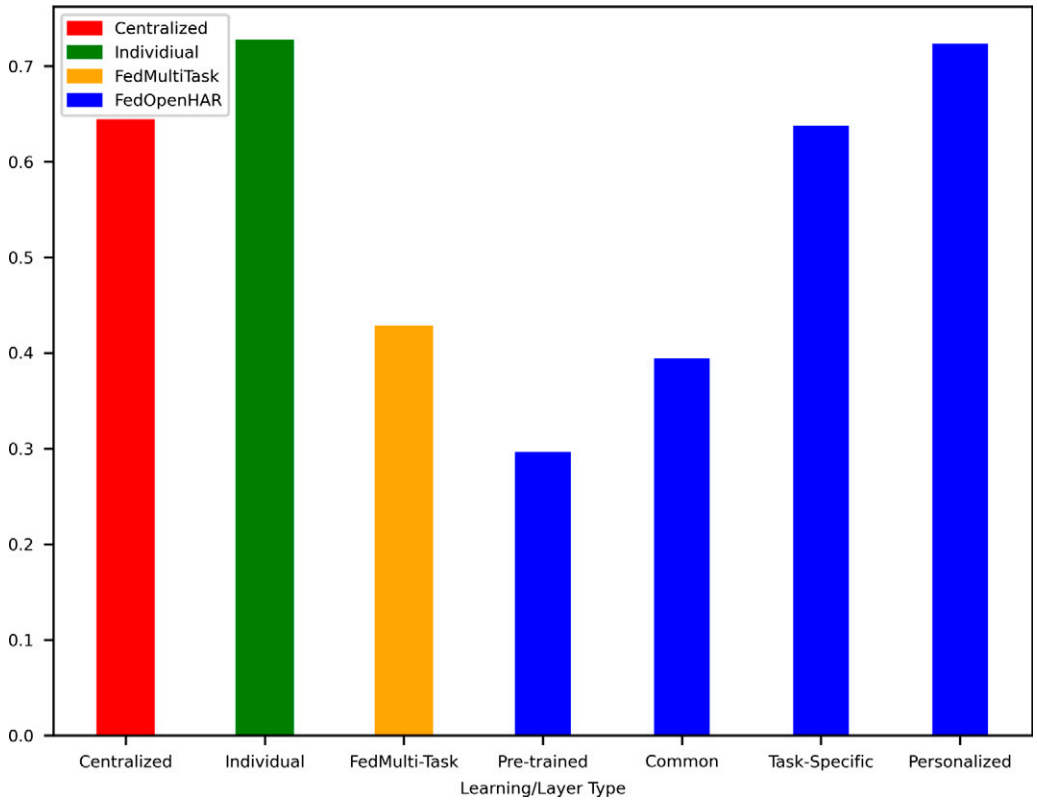

FIG. 9. Weighted average accuracies of clients by layers and comparison with the individual experiments' average.

5.6. Impact of the federated averaging algorithm

As mentioned, this study's default federated aggregation algorithm is FedAVG (McMahan et al., 2017). However, there are also different algorithms for federation. For comparisons, we changed the federation algorithm to FedProx (Li et al., 2020) and FedYogi (Reddi et al., 2021). FedProx adds a regularization parameter (μ) to the FedAVG, and various regularization values are explored in this section. For FedYogi, the federated averaging algorithm uses the federated version of the centralized Yogi optimizer.

As seen in Table 5, regularization with FedProx with $\mu = 0$ equals the FedAVG method, which is weighted averaging without any regularization parameter, and the other regularization values do not have a major effect on the overall performance. FedYogi performs the worst among all other federated averaging algorithms, and FedAVG still fits best in all of the federated averaging algorithms. FedAVG is one of the most common averaging algorithms used in the literature and consists of basic weighted averaging for aggregation. However, FedProx adds one regularization parameter, and FedYogi adds four different parameters to the aggregation. The variation of the algorithm and those parameters did not improve the results, but a more simplistic FedAVG achieved similar or better results. This is due to the fact that the whole FedOpenHAR algorithm consists of various little steps, and each of them contains multiple averaging of results (see success metrics in Section 5.1). This much averaging was not compatible with the more advanced aggregation methods.

6. CONCLUSION AND FUTURE STUDIES

This article introduces FedOpenHAR, a novel federated multitask transfer learning framework for human activity classification and device position identification using motion sensor data. Its performance is compared against various centralized learning models and federated learning approaches based on the DeepConvLSTM architecture. The experiments are conducted using the integrated OpenHAR dataset, which consolidates ten smaller datasets.

In the multitask federated transfer learning setup, FedOpenHAR achieved a success rate of 72.4%, closely matching the baseline of individual training (72.6%) and significantly outperforming the fully centralized approach (64.5%). Among the federated algorithms tested, FedAVG demonstrated the best performance. Its simple yet effective weight-averaging strategy proved highly compatible with the deep hierarchical architecture of FedOpenHAR, maximizing its potential.

The study opens a new viewpoint to federated HAR research, as it is being the first implementation of HAR using a federated multitask transfer learning architecture. However, a noteworthy observation is the minimal difference between the success rates of FedOpenHAR and the baseline. This can be attributed to the multi-layered nature of the architecture, where each layer undergoes numerous training epochs using the multi-layered DeepConvLSTM network. Consequently, changes to specific parameters have limited impact on the overall success, as their effects remain local and do not significantly influence the final results, due to the multiple aggregations of model parameters, as well as multiple averaging in success metrics.

While the default settings performed adequately, there are still opportunities for further investigation. Future studies could explore the impact of client selection strategies, such as prioritizing larger clients, and simulate scenarios involving random client drops or delayed clients. Furthermore, evaluating how these factors interact

TABLE 5. WEIGHTED ACCURACIES FOR VARIOUS FEDERATED AVERAGING ALGORITHMS[a]

	Federated averaging algorithm				
Layers	*FedAVG*	*FedProx u = 0.1*	*FedProx u = 0.3*	*FedProx u = 0.6*	*FedYogi*
Pretrained	29.6%	29.6%	29.6%	29.6%	29.6%
Common	39.4%	38.6%	36.8%	40.9%	34.6%
Task-specific	63.8%	64.0%	64.1%	64.1%	47.5%
Personalized	**72.4%**	72.2%	72.3%	72.2%	66.4%

[a]The baseline accuracy with individual experiments is 72.6%.

with parameters such as latency and resource usage could yield deeper insights into optimizing federated learning architectures.

AUTHORS' CONTRIBUTIONS

E.İ.: Methodology, software, formal analysis, investigation, data curation, writing—original draft, writing—review and editing, visualization. O.D.İ.: Conceptualization, resources, validation, writing—original draft, writing—review and editing, supervision, project administration.

AUTHOR DISCLOSURE STATEMENT

The authors declare that they have no conflicting financial interests.

FUNDING INFORMATION

This study is not funded by any government, company, or foundation.

REFERENCES

Anguita D, Ghio A, Oneto L, et al. A public domain dataset for human activity recognition using smartphones. In: International European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning, Bruges, Belgium; 2013; pp. 437–442.

Arivazhagan MG, Aggarwal V, Singh AK, et al. Federated learning with personalization layers. arXiv:1912.00818 2019.

Bock M, Hölzemann A, Moeller M, et al. Improving deep learning for har with shallow lstms. In: International Symposium on Wearable Computers, New York, USA; 2021; pp. 7–12.

Bursa SO, Incel OD, Isiklar Alptekin G. Personalized and motion-based human activity recognition with transfer learning and compressed deep learning models. Computers and Electrical Engineering 2023;109(2):108777.

Chen Y, Qin X, Wang J, et al. Fedhealth: A federated transfer learning framework for wearable healthcare. IEEE Intell Syst 2020;35(4):83–93.

Ek S, Portet F, Lalanda P, et al. Evaluation of federated learning aggregation algorithms: Application to human activity recognition. In: International Joint Conference on Pervasive and Ubiquitous Computing and International Symposium on Wearable Computers, New York, USA; 2020; pp. 638–643.

Kalabakov S, Jovanovski B, Denkovski D, et al. Federated learning for activity recognition: A system level perspective. IEEE Access 2023;11(1):64442–64457.

Keçeci C, Shaqfeh M, Mbayed H, et al. Multi-task and transfer learning for federated learning applications. arXiv:2207.08147 2022.

Kumar P, Chauhan S, Awasthi LK. Human activity recognition (har) using deep learning: Review, methodologies, progress and future research directions. Arch Computat Methods Eng 2024;31(1):179–219.

McMahan HB, Moore E, Ramage D, et al. A. Communication-efficient learning of deep networks from decentralized data. arXiv:1602.05629 2017.

Murahari VS, Plötz T. On attention models for human activity recognition. In: International Symposium on Wearable Computers, New York, USA; 2018; pp 100–103.

Ordóñez FJ, Roggen D. Deep convolutional and lstm recurrent neural networks for multimodal wearable activity recognition. Sensors 2016;16(1):115.

Ouyang X, Xie Z, Zhou J, et al. Clusterfl: A similarity-aware federated learning system for human activity recognition. In: International Conference on Mobile Systems, Applications, and Services, New York, USA; 2021; pp. 54–66.

Presotto R, Civitarese G, Bettini C. Fedclar: Federated clustering for personalized sensor-based human activity recognition. In: International Conference on Pervasive Computing and Communications, Pisa, Italy; 2022; pp. 227–236.

Sashank R, Zachary C, Manzil Z, et al. Adaptive Federated Optimization. arxiv:2003.00295 2021.

Siirtola P, Koskimäki H, Röning J. Openhar: A matlab toolbox for easy access to publicly open human activity data sets. In: International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers, New York, USA; 2018;pp. 1396–1403.

Smith V, Chiang C-K, Sanjabi M, et al. Federated multi-task learning. arXiv:1602.05629 2018.

Tan AZ, Yu H, Cui L, et al. Towards personalized federated learning. IEEE Trans Neural Netw Learn Syst 2023; 34(12):9587–9603.

Tian L, Kumar SA, Manzil Z, et al. Federated Optimization in Heterogeneous Networks. arxiv:1812.06127 2020.

Truong N, Sun K, Wang S, et al. Privacy preservation in federated learning: An insightful survey from the gdpr perspective. arXiv:2011.05411 2021.

Wu Q, He K, Chen X. Personalized federated learning for intelligent iot applications: A cloud-edge based framework. IEEE Open J Comput Soc 2020;1(1):35–44.

Yang Q, Liu Y, Chen T, et al. Federated machine learning: Concept and applications. arXiv:1902.04885 2019.

Address correspondence to:
Mr. Egemen İşgüder
Pervasive Systems Research Group
Faculty of EEMCS
University of Twente
Enschede
The Netherlands

E-mail: egemen.isguder@utwente.nl

Dr. Özlem Durmaz İncel
Pervasive Systems Research Group
Faculty of EEMCS
University of Twente
Enschede
The Netherlands

E-mail: ozlem.durmaz@utwente.nl